

# Stat 435 Lecture Notes 6

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## Topics to discuss

- Multinomial regression
- Poisson regression

## Multinomial regression

### Observations in classes

Each observations  $y_i, i = 1, \dots, n$  belongs to one and only one of  $K$  classes:

- Class 1 obs:  $\tilde{y}_{11}, \tilde{y}_{12}, \dots, \tilde{y}_{1,n_1}$

- Class 2 obs:  $\tilde{y}_{21}, \tilde{y}_{22}, \dots, \tilde{y}_{2, n_2}$
- Class ... obs
- Class  $K$  obs:  $\tilde{y}_{K1}, \tilde{y}_{K2}, \dots, \tilde{y}_{K, n_K}$

## Probability mass function

- Model:  $P(y_i = k) = p_{ik}$
- Equivalent formulation: set  $y_{ij} = 1$  if and only if  $y_i = j$ . Then *probability mass function (pmf)*

$$P(y_i = k) = p_{i1}^{y_{i1}} \cdots p_{ik}^{y_{ik}} \cdots p_{iK}^{y_{iK}} = \prod_{j=1}^K p_{ij}^{y_{ij}}$$

- Example with  $K = 3$ ; comparison with Bernoulli setting
- When  $\{y_i\}_{i=1}^n$  are independent, their joint likelihood is

$$L = \prod_{i=1}^n p_{i1}^{y_{i1}} \cdots p_{ik}^{y_{ik}} \cdots p_{iK}^{y_{iK}} = \prod_{i=1}^n \prod_{j=1}^K p_{ij}^{y_{ij}}$$

## Model

- $Y \in \{1, 2, \dots, K\}$  such that

$$\sum_{k=1}^K \Pr(Y = k) = 1 \text{ and } \prod_{k=1}^K \Pr(Y = k) > 0$$

- Model  $\Pr(Y = k)$  by  $X = (X_1, X_2, \dots, X_p)^T$ 
  - Set  $p_k(X) = \Pr(Y = k|X)$  for  $k = 1, \dots, K$
  - For  $k = 1, \dots, K - 1$ , set

$$\log \left( \frac{p_k(X)}{p_K(X)} \right) = \beta_{k0} + \beta_k^T X, \quad (1)$$

$$\text{where } \beta_k = (\beta_{k1}, \beta_{k2}, \dots, \beta_{kp})^T$$

## Equivalent model

Model is equivalent to:

$$p_k(x) = \Pr(Y = k|X = x) = \frac{\beta_{k0} + \beta_k^T x}{1 + \sum_{l=1}^{K-1} \exp(\beta_{l0} + \beta_l^T x)}$$

for  $k = 1, \dots, K - 1$  and

$$p_K(x) = \Pr(Y = K|X = x) = \frac{1}{1 + \sum_{l=1}^{K-1} \exp(\beta_{l0} + \beta_l^T x)}$$

## Model: sample version

- The modeling strategy

$$\log \left( \frac{p_k(x_i)}{p_K(x_i)} \right) = \beta_{k0} + \beta_k^T x_i$$

is equivalent to the formulation provided by **glmnet** as

$$P(y_i = k | X = x_i) = \frac{\exp(\tilde{\beta}_{k0} + \tilde{\beta}_k^T x_i)}{\sum_{l=1}^K \exp(\tilde{\beta}_{l0} + \tilde{\beta}_l^T x_i)}$$

## Check on model

- **glmnet** fits model:

$$P(Y = k | X = x) = \frac{\exp(\beta_{k0} + \beta_k^T x)}{\sum_{i=1}^K \exp(\beta_{i0} + \beta_i^T x)} \text{ for } k = 1, \dots, K$$

- To check the linearity assumption, i.e.,

$$\log \frac{P(Y = k | X = x)}{P(Y = K | X = x)} = \exp(\beta_{k0} + \beta_k^T x) \text{ for } k = 1, \dots, K - 1$$

we need the estimated coefficients for each class probability  $P(Y = k | X = x)$

- In general, a goodness-of-fit test for the multinomial logistic regression model can be the Pearson's chi-square test or the likelihood ratio test

## Check on model

Checking if the multinomial logistic regression assumption is adequate:

- Under suitable conditions,  $E(y_i - \hat{\mu}_i) \rightarrow 0$  as sample size  $n \rightarrow \infty$  when no regularization that induces bias in  $\hat{\mu}_i$  is used in model fitting, where  $\hat{\mu}_i$  is the estimated mean for  $y_i$
- When LASSO is used, biased is often induced to  $\hat{\mu}_i$  as an estimate of  $E(y_i)$ , and in this case we can only have  $E(y_i - \hat{\mu}_i) \approx 0$  as  $n \rightarrow \infty$

## Poisson regression

### Poisson random variable

- A Poisson random variable  $Y$  is defined via its *pmf* as

$$\Pr(Y = k) = \frac{\mu^k}{k!} e^{-\mu} \text{ for } k \in \{0, 1, 2, \dots\}$$

where  $\mu > 0$  is fixed. So,  $E(Y) = V(Y) = \mu$ .

- Given observations  $(y_i, x_i), i = 1, \dots, n$ , set

$$p(x_i) = E(y_i | x_i) = \mu_i$$

- *Poisson linear regression* model:

$$\log(p(x_i)) = \beta_0 + \beta_1 x_i$$

- Above model is equivalent to

$$\mu_i = p(x_i) = \exp(\beta_0 + \beta_1 x_i)$$

## Model assumptions

- Each  $y_i$  is a Poisson random variable with mean  $\mu_i$
- $\log(p(x_i))$  is a linear function of  $x_i$  for  $1 \leq i \leq n$
- The pairs  $\{(x_i, y_i)\}_{i=1}^n$  are independent
- Joint likelihood function is  $L(\beta) = \prod_{i=1}^n \frac{\mu_i^{y_i}}{y_i!} e^{-\mu_i}$  and

$$\log L(\beta) = \sum_{i=1}^n [-\exp(\beta_0 + \beta_1 x_i)] + \sum_{i=1}^n [y_i (\beta_0 + \beta_1 x_i) - \log(y_i!)]$$

with coefficients  $\beta = (\beta_0, \beta_1)$

## Estimated model

- *Maximum likelihood estimate (MLE)*  $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1)$  of  $\beta = (\beta_0, \beta_1)$  satisfies  $\hat{\beta} \in \operatorname{argmin}_{\beta} L(\beta)$
- Predictions on  $y_i$  can be made as

$$\Pr(y_i = k | x_i) = \frac{\hat{\mu}_i^k}{k!} e^{-\hat{\mu}_i} \text{ with } \hat{\mu}_i = \exp(\hat{\beta}_0 + \hat{\beta}_1 x_i)$$

- Namely,  $y_i$  is predicted by  $\hat{y}_i$  for which

$$\Pr(\hat{y}_i = k | x_i) = \frac{\hat{\mu}_i^k}{k!} e^{-\hat{\mu}_i}$$

## Model diagnosis

- *Reduced model*  $\mu_i = p(x_i) = \exp(\beta_0 + \beta_1 x_i)$  versus *Full model*  $E(y_i) = \mu_i$
- Model deviance is

$$D = -2 \times [\ln L(\text{Reduced model}) - \ln L(\text{Full model})] = -2 \left[ \sum_{i=1}^n y_i \ln \frac{\hat{\mu}_i}{y_i} + \sum_{i=1}^n (y_i - \hat{\mu}_i) \right] = \sum_{i=1}^n d_i^2$$

## Model diagnosis

- Deviance residual

$$d_i = \operatorname{sgn}(y_i - \hat{\mu}_i) \sqrt{-2y_i \ln \frac{\hat{\mu}_i}{y_i} - (y_i - \hat{\mu}_i)}$$

- Pearson residual

$$r_i = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$$

and Pearson chi-square goodness of fit statistic

$$\xi^2 = \sum_{i=1}^n \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i} = \sum_{i=1}^n r_i^2$$

## Model diagnosis

- Studentized Pearson residual

$$\tilde{r}_i = \frac{r_i}{\sqrt{1 - h_{ii}}},$$

where

- $h_{ii}$  is the  $i$ -th diagonal entry of

$$H = W^{1/2} X (X' W X)^{-1} X' W^{1/2}$$

- $W$  is the  $n \times n$  diagonal matrix with entries  $\hat{\mu}_i$
- $X$  is the  $n \times p$  design matrix

## Model diagnosis

- Poisson model is correct  $\implies E(y_i - \hat{\mu}_i) = 0$  asymptotically
- Check linearity assumption: plot  $\ln \hat{\mu}_i$  against  $x_i$
- Check Poisson model adequacy: plot  $r_i$  against  $\hat{\mu}_i$
- Check for influential observations: plot *delta chi-square statistic*

$$\Delta \xi_{(i)}^2 = \xi^2 - \xi_{(i)}^2$$

and *delta deviance statistic*

$$\Delta D_{(i)} = D - D_{(i)}$$

against  $i$  (or against  $\hat{\mu}_i$  or  $\hat{\eta}_i$ )

## Example: data

Miller Lumber Company survey data:

- $X_1$ : number of housing units
- $X_2$ : average income, in dollars
- $X_3$ : average housing unit age, in years
- $X_4$ : distance to nearest competitor, in miles
- $X_5$ : distance to store, in miles
- $Y$ : number of customers who visited store from census tract

(Credit: authors of book “Applied linear statistical models”)

Example: partial data

**TABLE 14.14**  
Data—Miller  
Lumber  
Company  
Example.

| Census Tract<br><i>i</i> | Housing Units<br>$X_1$ | Average Income<br>$X_2$ | Average Age<br>$X_3$ | Competitor Distance<br>$X_4$ | Store Distance<br>$X_5$ | Number of Customers<br>$Y$ |
|--------------------------|------------------------|-------------------------|----------------------|------------------------------|-------------------------|----------------------------|
| 1                        | 606                    | 41,393                  | 3                    | 3.04                         | 6.32                    | 9                          |
| 2                        | 641                    | 23,635                  | 18                   | 1.95                         | 8.89                    | 6                          |
| 3                        | 505                    | 55,475                  | 27                   | 6.54                         | 2.05                    | 28                         |
| ...                      | ...                    | ...                     | ...                  | ...                          | ...                     | ...                        |
| 108                      | 817                    | 54,429                  | 47                   | 1.90                         | 9.90                    | 6                          |
| 109                      | 268                    | 34,022                  | 54                   | 1.20                         | 9.51                    | 4                          |
| 110                      | 519                    | 52,850                  | 43                   | 2.92                         | 8.62                    | 6                          |

Example: fitted model

**TABLE 14.15**  
Fitted Poisson  
Response  
Function and  
Related  
Results—  
Miller Lumber  
Company  
Example.

| (a) Fitted Poisson Response Function                                                 |                                  |                              |       |         |
|--------------------------------------------------------------------------------------|----------------------------------|------------------------------|-------|---------|
| $\hat{\mu} = \exp[2.942 + .000606X_1 - .0000117X_2 - .00373X_3 + .168X_4 - .129X_5]$ |                                  |                              |       |         |
| $DEV(X_0, X_1, X_2, X_3, X_4, X_5) = 114.985$                                        |                                  |                              |       |         |
| (b) Estimated Coefficients, Standard Deviations, and $G^2$ Test Statistics           |                                  |                              |       |         |
| Regression Coefficient                                                               | Estimated Regression Coefficient | Estimated Standard Deviation | $G^2$ | P-value |
| $\beta_0$                                                                            | 2.9424                           | .207                         |       |         |
| $\beta_1$                                                                            | .0006058                         | .00014                       | 18.21 | .000    |
| $\beta_2$                                                                            | -.00001169                       | .0000021                     | 31.80 | .000    |
| $\beta_3$                                                                            | -.003726                         | .0018                        | 4.38  | .036    |
| $\beta_4$                                                                            | .1684                            | .026                         | 41.66 | .000    |
| $\beta_5$                                                                            | -.1288                           | .016                         | 67.50 | .000    |

- $G^2 = -2 \ln[L(R)/L(R)] \sim$  chi-square distribution

Example: diagnosis

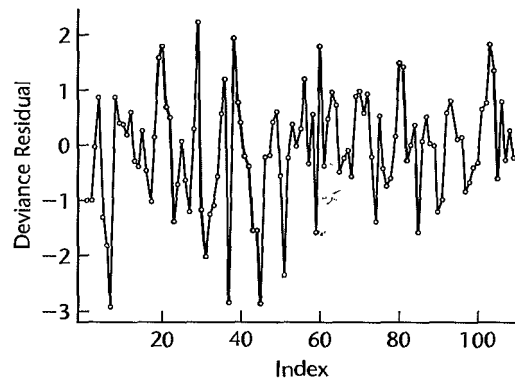
**TABLE 14.16**  
Responses,  
Fitted Values,  
and Deviance  
Residuals—  
Miller Lumber  
Company  
Example.

| Census Tract<br><i>i</i> | $Y_i$ | $\hat{\mu}_i$ | $dev_i$ |
|--------------------------|-------|---------------|---------|
| 1                        | 9     | 12.3          | -.999   |
| 2                        | 6     | 8.8           | -.992   |
| 3                        | 28    | 28.1          | -.024   |
| ...                      | ...   | ...           | ...     |
| 108                      | 6     | 5.3           | .289    |
| 109                      | 4     | 4.4           | -.197   |
| 110                      | 6     | 6.4           | -.171   |

- deviance residual vs fitted value; table shows no serious issue

## Example: diagnosis

**FIGURE 14.20**  
Index Plot of  
Deviance  
Residuals—  
Miller Lumber  
Company  
Example.



- a few large negative deviance residuals associated tracts with zero customers; hard to fit these values

## License and session Information

License

```
> sessionInfo()
R version 3.5.0 (2018-04-23)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 10 x64 (build 19044)
```

Matrix products: default

locale:

```
[1] LC_COLLATE=English_United States.1252
[2] LC_CTYPE=English_United States.1252
[3] LC_MONETARY=English_United States.1252
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.1252
```

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods
[7] base
```

other attached packages:

```
[1] knitr_1.21
```

loaded via a namespace (and not attached):

```
[1] compiler_3.5.0  magrittr_1.5    tools_3.5.0
[4] htmltools_0.3.6 yaml_2.2.0      Rcpp_1.0.12
[7] stringi_1.2.4   rmarkdown_1.11 stringr_1.3.1
[10] xfun_0.4        digest_0.6.18   evaluate_0.13
```